

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0892

Paloma 100 kV
October 03, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0892. The Interconnection Customer proposes a 287.1 MW wind facility interconnecting at the Paloma 100 kV substation in the Illinois area. The request was received on November 14, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 280 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$60,719,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
EASY VLY – HEFFLEY TAP 100 KV	WALDEN – W AMESBURY 100 KV line	293.0	119	80.3
PALOMA – WALDEN 100 KV	HEFFLEYTAP – EASY VLY 100 KV	236.7	139	100.0
EASY VLY – SHADY OAKS WIND FARM 100 KV	WALDEN – WEST AMESBURY 100 KV	203.9	116	60.9
PALOMA – HEFFLEY TAP 100 KV	WADEN – W AMESBURY 100 KV	200.1	146	80.3
WALDEN – W AMESBURY 100 KV	TAYLORVILLE SOUTH 230/1 KV TX	127.2	306	47.8
EASY VLY – CANTON 100 KV line	EASY VLY – SHADY OAKS WIND FARM 100 KV	118.4	125	34.8
OKAHUMPKA – ARKANSAS 100 KV Line	SHADY OAKS WINDFARM – EASY VLY 100 KV	118.1	111	34.8
KETTLE MORRAINE – CANTON 100 KV	EASY VLY – SHADY OAKS WIND FARM SUBSTATION 100 KV	118.0	121	34.8
KETTLE MORRAINE – OKAHUMPKA 100 KV	EASY VLY SUBSTATION – SHADY OAKS WIND FARM 100 KV	117.6	124	34.8
EXIRA – RUTHERFORD FARM 100 KV Line 1	RUTHERFORD FARM SUBSTATION – EXIRA 100 KV Circuit 2	108.1	144	5.3
EXIRA SUBSTATION – RUTHERFORD FARM 100 KV Circuit 2	EXIRA – RUTHERFORD FARM SUBSTATION 100 KV Circuit 1	108.1	144	5.3
ARKANSAS – YKK USA CHESTNEY 100 KV	EASY VLY – SHADY OAKS WIND FARM 100 KV Line	107.4	99	30.4
EXIRA – SHADY OAKS WINDFARM 100 KV Cir. 1	SHADY OAKS WIND FARM – EXIRA 100 KV Cir. 2	106.2	176	6.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CANAL RD – WEEDPATCH 230 KV	TREGO – BEAVER 500 KV L	105.4	966	7.6
POST ROCK – NEWPORT TIE 230 KV	TREGO – BEAVER 500 KV L	101.3	1,691	13.2
JONES CHAPEL 500-230 kV Autotransformer	DEKALB – JOES CHAPEL 500 KV	101.1	1,864	36.4
SIDNEY A MURRAY JR HYDROELECTRIC SUBSTATION – MINNEHAHA 230 KV	DEKALB – RICHLAND 500 KV	101.0	1,138	16.4
SHADY OAKS WIND FARM – ALDER SUBSTATION 230 KV	SHADY OAKS WIND FARM – ALDER 500 KV	100.5	1,011	25.3
MCNEIL – BREED 230 KV L	DEKALB – RICHLAND 500 KV	100.4	1,669	25.6
SIDNEY A MURRAY JR HYDROELECTRIC – BREED 230 KV	DEKALB – RICHLAND SUBSTATION 500 KV	100.4	1,669	25.6
POST ROCK – MYRTLE POINT 230 KV line	TREGO – BEAVER 500 KV	100.3	1,762	12.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
EASY VALLEY – HEFFLEY TAP 100 KV	Rebuild line with higher-capacity conductor	\$14,371,000	83% (\$11,886,000)
PALOMA – WALDEN 100 KV	Rebuild line with higher-capacity conductor	\$13,710,000	100% (\$13,710,000)
EASY VLY – SHADY OAKS WIND FARM 100 KV	Rebuild line with higher-capacity conductor	\$5,400,000	76% (\$4,086,000)
PALOMA – HEFFLEY TAP 100 KV	Rebuild line with higher-capacity conductor	\$4,519,000	83% (\$3,737,000)
WALDEN – W AMESBURY 100 KV	Rebuild line with higher-capacity conductor	\$3,978,000	39% (\$1,536,000)
EASY VLY – CANTON 100 KV line	Rebuild line with higher-capacity conductor	\$1,851,000	100% (\$1,851,000)
ARKANSAS – OKAHUMPKA SUBSTATION 100 KV	Rebuild line with higher-capacity conductor	\$1,941,000	100% (\$1,941,000)
KETTLE MORRAINE – CANTON 100 KV	Rebuild line with higher-capacity conductor	\$8,744,000	100% (\$8,744,000)
OKAHUMPKA – KETTLE MORRAINE 100 KV Line	Rebuild line with higher-capacity conductor	\$4,107,000	100% (\$4,107,000)
EXIRA – RUTHERFORD FARM 100 KV Line 1	Reconductor line	\$3,184,000	38% (\$1,206,000)
EXIRA SUBSTATION – RUTHERFORD FARM 100 KV Circuit 2	Reconductor line	\$4,708,000	38% (\$1,784,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ARKANSAS – YKK USA CHESTNEY 100 KV	Reconductor line	\$379,000	100% (\$379,000)
EXIRA – SHADY OAKS WIND FARM 100 KV	Reconductor line	\$938,000	38% (\$355,000)
CANAL RD – WEEDPATCH 230 KV	Reconductor line	\$7,222,000	3% (\$194,000)
POST ROCK – NEWPORT TIE 230 KV	Reconductor line	\$11,781,000	3% (\$317,000)
JONES CHAPEL 500-230 kV Autotransformer	Replace transformer with higher-rated unit	\$17,246,000	3% (\$466,000)
MINNEHAHA – SIDNEY A MURRAY JR HYDROELECTRIC 230 KV	Reconductor line	\$13,843,000	3% (\$373,000)
SHADY OAKS WIND FARM – ALDER SUBSTATION 230 KV	Reconductor line	\$10,458,000	30% (\$3,182,000)
MCNEIL – BREED 230 KV L	Reconductor line	\$7,223,000	3% (\$194,000)
SIDNEY A MURRAY JR HYDROELECTRIC – BREED 230 KV	Reconductor line	\$18,212,000	3% (\$490,000)
POST ROCK – MYRTLE PT 230 KV line	Reconductor line	\$6,743,000	3% (\$181,000)
	Total allocated to project		\$60,719,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.